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A NEW COMBINATION IN *SANICULA CRASSICAULIS* (APIACEAE)

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ABSTRACT

During the course of a floristic study of the Cal Poly Swanton Pacific Ranch, in Santa Cruz County, we observed and collected several specimens that keyed to *Sanicula crassicaulis* Poepp. ex. D.C. but were notably distinct from other *S. crassicaulis* that we collected in the area. On further investigation, we found that these plants match the type specimen of *S. nudicaulis* Hook & Arn., described in 1839 and later synonymized with *S. crassicaulis*. This taxon is distinguished primarily by its long, sinuate marginal leaf trichomes, in contrast to the short, straight marginal trichomes of *S. crassicaulis*. Additionally, the length of the most proximal prickles on the schizocarps is about the same as the length of the most distal rather than the distal prickles being much longer as in *S. crassicaulis* and the angle of attachment of the prickles is more or less perpendicular as opposed to acute in *S. crassicaulis*. The known range of this taxon is between the San Francisco Bay Area and Santa Barbara County, generally near the coast. Here we recognize this taxon as a variety in *S. crassicaulis*. We provide a diagnosis, a map of the known range of the taxon and a key to the varieties of *S. crassicaulis* in California.

Key Words: new combination, *Sanicula menziesii* var. *nudicaulis*, Sanicula nudicaulis, Swanton, trichomes, schizocarp

Sanicula crassicaulis Poepp. ex D.C. is an herbaceous perennial species distributed throughout western North America from Vancouver Island to northern Baja California and reappearing in central Chile. This species has long been recognized as morphologically highly variable and can be found in open grasslands, shrublands, woodlands, riparian corridors, and mixed conifer forest. Here we argue that the taxon *S. nudicaulis* Hook. & Arn., formerly treated as a synonym of *S. crassicaulis* var. *crassicaulis,* should be recognized as a variety within *S. crassicaulis*.

The name *Sanicula crassicaulis,* was published in 1830 from specimens collected in Chile by Poepigg and therefore has taxonomic priority as the valid name for the species (de Candolle 1830). North American specimens, now treated as *S. crassicaulis,* were originally described as two separate species. The names *Sanicula menziesii* Hook. & Arn. and *Sanicula nudicaulis* Hook & Arn. were published in 1839 from specimens collected in California in 1827, during Captain Beechey's Voyage of 1825-1828 (Arnott et al. 1839). The name *S. menziesii* referred to plants that were largely indistinguishable from *S. crassicaulis* aside from their distribution. The name *S. nudicaulis* referred to plants with a caulescent stem, leaves cordate in three parts and setose leaf margins (Fig. 1).

Willis Lynn Jepson treated *S. nudicaulis* as a variety of *S. menziesii* (*S. menziesii* Hook & Arn. var. *nudicaulis* Jeps.) in his 1911 Flora of Western Middle California (Jepson 1911). In 1923, he published two new varieties of *S. menziesii* in Madroño: var. *foliacea* Jeps. and var. *pedata* Jeps. Jepson separated the varieties in this way: plants with thin, only partly divided leaves and with leaf segments that are less lobed, fall into var. *nudicaulis*; stout plants with thicker leaves that are pedately divided into cuneate segments with serrate, bristle tipped margins fall into var. *pedata*; and low growing, leafy plants, with bracteal leaves that are notably larger fall into var. *foliacea* (Jepson 1923). The types for all of these taxa were collected in California.

In 1913, Herman Wolff placed *S. menziesii* into *Sanicula crassicaulis* as a variety, resulting in the combination *S. crassicaulis* Poepp. ex. D.C. var. *menziesii* H.Wolff, based off of the fact that he could find no consistent morphological differences between North and South American plants. He listed *S. nudicaulis* Hook & Arn. as a synonym for this new combination (Wolff 1913). In 1936, Mildred Mathias treated *S. menziesii* as a synonym to *S. crassicaulis,* stating, "these two entities and their various segregates represent a variable species." This resulted in *S. nudicaulis, S. menziesii* var. *nudicaulis*, and *S. crassicaulis* var. *menziesii* all being treated as

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FIG. 1. Type specimen of *Sanicula nudicaulis*. The inset shows 16 times magnification of the leaf marginal trichomes.

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FIG. 2. Known range of *Sanicula crassicaulis* var. *nudicaulis*. Points represent herbarium specimens or seed collections. Map datum is WGS84 and the coordinate system is UTM.

synonyms for *S. crassicaulis* (Mathias 1936). This taxonomy was largely followed by Shan and Constance in their 1951 monograph of the genus (Shan and Constance 1951) and by Ritchie Bell in his 1954 study of polyploidy and putative hybridization in *S. crassicaulis* (Bell 1954). Molecular phylogenetic work has confirmed that Chilean and North American *S. crassicaulis* do form a monophyletic group (Vargas et al. 1999). Evidence suggests that this amphitropic distribution is due to long distance dispersal, likely bird mediated, from North to South America (Vargas et al. 1998). As *S. crassicaulis* in North America hosts three ploidy levels, as well as a high level of morphological diversity, we would expect individuals in South America to show a lesser degree of variation. This is because of the likely bottleneck effect resulting from a long distance dispersal event (Vargas et al. 1998). For this reason, we do not expect to see the same stable forms occurring in Chile as in California, but we have taken steps to examine the available material from South America.

During a floristic study of the Cal Poly Swanton Pacific Ranch, near Davenport, CA, we encountered several populations of a distinct form of *Sanicula crassicaulis* with no evidence of intermediate forms. Upon further investigation, we found several populations of the distinct form, occurring between the San Francisco Bay Area and Santa Barbara County (Fig. 2). We reviewed many herbarium specimens, as well as living material, and have concluded that some plants in California share key characteristics with the type specimen of *S. nudicaulis* and are morphologically distinct from *S. crassicaulis* var. *crassicaulis,* deserving recognition as a variety of *S. crassicaulis*.

Previously, this taxon was distinguished by having caulescent stems and leaves that are somewhat less lobed, thinner, and with much longer, undulate, marginal trichomes than *Sanicula crassicaulis* var.

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crassicaulis. We found that the long, undulate marginal trichomes on fresh leaves can be used to reliably distinguish this taxon and agree with previous treatments that the leaves tend to be thinner, although we have not measured this trait. We have observed many different stem branching patterns in this taxon and so do not believe that this taxon has markedly caulescent stems. We have observed a pattern of leaves that are generally less divided, but we do not believe this is a character that can be used to readily separate this taxon from others. In addition to the leaf traits used to distinguish this taxon, we add two fruit characters, the angle of attachment of the prickles to the schizocarps and the difference in length between the most proximal and the most distal prickles. We choose to recognize this taxon as a variety, as there are already two existing varieties in *S. crassicaulis, S. crassicaulis* Poepp ex. D.C. var. *crassicaulis* and *S. crassicaulis* Poepp ex. D.C. var. *tripartita* H.Wolff.

Methods

We measured a total of 75 accessions to examine variation in fruit morphological characters. Twenty-six of these were fruit accessions banked at the UCSC Arboretum seed bank, 39 were herbarium specimens from the UC Davis Herbarium (DAV), UC Berkeley and Jepson Herbaria (UC/JEPS), four were images from Royal Botanic Garden Edinburg (E) and four were fruit and herbarium collections collected by the authors and vouchered at the Hoover Herbarium (OBI). The geographic range of *S. crassicaulis* var. *crassicaulis* specimens measured covers most of the range in California, as well as a small sample from South America. We measured *S. crassicaulis* var. *nudicaulis* specimens collected from the San Francisco Bay Area to Santa Barbara County. These covered the current known range of the taxon (Fig. 2), although samples were concentrated from the Davenport area and from San Luis Obispo County.

We measured morphological variation in two fruit traits. These traits are 1) the angle of attachment of the prickles on the schizocarps, as well as 2) the difference in length between the most distal and the most proximal prickles. Up to 10 schizocarps were measured per accession or herbarium specimen. Specimens and seed collections were included only if these traits could be measured. Specimens in strictly vegetative condition were excluded. Fruits were photographed using a Leica EZ4 dissecting scope with an integrated digital camera and imported into ImageJ (Rueden et al. 2017). Schizocarps were imaged with the adaxial face facing upward. The abaxial face was supported on a bed of Velcro so that the adaxial face was as close to perpendicular with the camera lens as possible. This allowed us to capture images where the first series of prickles are seen in true length (Fig. 3C, 3D). For measurements made on images of Chilean specimens, fruits were selected that were positioned on the sheet to allow true length measurements. On each schizocarp, we measured the length and angle of attachment of the most distal, the most proximal, and one medial prickle closest to the adaxial face of the schizocarp (Fig. 3C). The angle of attachment was measured as the angle between the line running from the tip to the base of the prickle and the line tangent to the curve of the margin of the schizocarp (Fig. 3D). Angle of attachment was not measured on images of Chilean specimens. Prickle graduation is the length of the most distal prickle divided by the most proximal. Per fruit, the angle of prickle attachment is the average of the angle of attachment of the three prickles measured. Reported means, ranges and standard deviations were taken from the calculated per fruit values for angle of attachment and prickle graduation.

We measured leaf marginal trichome length on 111 individuals: 44 *S. crassicaulis* var. *crassicaulis* and 67 *S. crassicaulis* var. *nudicaulis*. We measured marginal trichome length on imaged specimens, as well as photographs of leaves in the field, due to the fact that the leaf marginal trichomes are often lost on herbarium specimens. Imaged specimens measured included the types for *Sanicula crassicaulis* var. *crassicaulis*. We photographed leaves alongside a ruler using a 10x macro lens (Moment macro lens V2, 25 mm) mounted on an iPhone 8 (Fig. 3A, 3B). Specimen images measured were collections from Chile and Argentina and held in the herbaria of Martin-Luther-Universität (HAL), Muséum National d'Histoire Naturelle (P), California Botanic Garden (RSA), Field Museum of Natural History (F), Royal Botanic Garden Edinburg (E), and Royal Botanic Garden (K). Accession numbers and the author's varietal determinations are listed in Appendix 1. All leaves measured in the field were from plants occurring more or less sympatrically in a population at Montaña de Oro State Park, San Luis Obispo, CA. We photographed one leaf per plant, and we chose leaves with intact marginal trichomes to photograph. We used ImageJ to measure the length of the marginal trichomes in the leaf photos. One trichome was haphazardly chosen to measure for each leaf photograph. The trichome was measured from the leaf margin to the tip of the trichome (Fig. 3A). Sinuate or bent trichomes were traced with the freehand line drawing tool in ImageJ (Fig. 3B). Reported means, ranges and standard deviations of leaf marginal trichome lengths represent the values measured for each variety.

Taxonomic Treatment

Sanicula crassicaulis Poepp. ex D.C. **var. *nudicaulis*** (Hook & Arn.) Kenny & West, **comb. nov.** Basionym: *Sanicula nudicaulis* Hook & Arn. Bot. Beechey Voy. (1839). Type: USA, CA. Monterey County. *David Douglas s.n.* (holotype: K000697304, isotype: E00369085).

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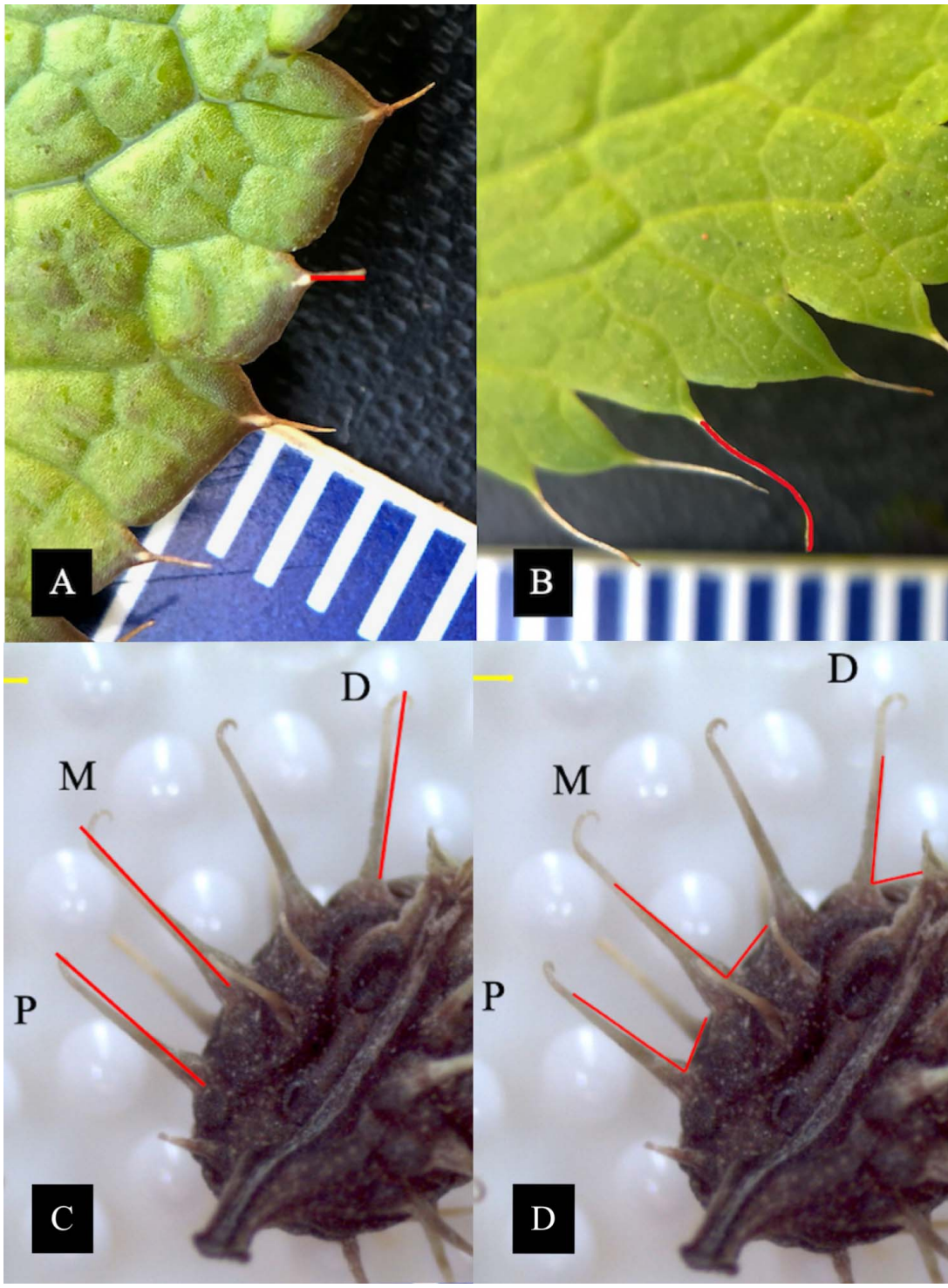

FIG. 3. Methods for measuring leaf trichome length, prickle length, and prickle angle of attachment. Image (A) shows an example of the method used to measure straight prickles as well as a close up of the marginal trichomes typical in var. *crassicaulis*. Image (B) shows the free hand method used to measure sinuate prickles, as well as a close up of the marginal trichomes typical in var. *nudicaulis*. In both images (C) and (D), P denotes the most proximal prickle, D the most distal, and M a medial prickle. Image (C) shows our method for measuring the length of the prickles on the schizocarp. The black lines represent the straight-line distance from the base of the prickle to the tip, which was measured in ImageJ. Image (B) shows the method for measuring the angle of attachment of the prickles. The angle formed by the black lines is the measured angle of attachment.

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Perennial, dying back to rosette in the winter. **Plants** stout, erect, 24–120 cm high, with a thin taproot. **Stems** solitary, fistulose, glabrous, generally erect, often many branched from base, but occasionally more or less caulescent to moderately branched distally. Stems when crushed have an acrid scent. **Leaves** basal and cauline, generally palmate to palmate ternate, green; blade 3–12 cm, generally rounded-cordate, lobes 3–5, obovate, generally shallowly cut, but can be + deeply cut, margins generally serrate to crenate, tipped with sinuate trichomes (2) 3–4 (7) mm in length. Marginal trichomes are often deciduous on older leaves and tend to be missing on herbarium specimens. Central lobe of leaf generally greater than 10 mm wide at base, distal segment 4–20 mm wide. Basal leaves generally long petioled. **Inflorescence**: peduncle 0.7–8 cm; bracts + 5, 1–2 mm, < heads, narrowly lanceolate; pedicels short. **Flower**: bisexual 3–8, staminate 3–5; calyx lobes fused at base, 0.5–0.7 mm, + lanceolate, acute; corolla yellow; styles 2× longer than calyx lobes. **Fruit**: 2–5 mm, + round, generally stalked, with stout prickles hooked at the tips. Prickles only slightly longer (~1.2×) distally than proximally, but proximal prickles occasionally little developed. Angle of attachment of prickles + perpendicular. **Seed**: face grooved.

Diagnosis: Leaves of *Sanicula crassicaulis* var. *nudicaulis* are recognizable in having leaf marginal trichomes that are (2) 3–4 (7) mm long and sinuate, in contrast to those of var. *crassicaulis* which are 1–2(3) mm long and straight. The marginal trichomes are often deciduous in age in var. *nudicaulis,* where those of var. *crassicaulis* persist in age or become partly deciduous, forming mucronate tips on the leaf serrations. Additionally, the prickles on the schizocarps are attached more or less perpendicularly, about 80° versus about 60° in var. *crassicaulis*. The most distal rank of prickles in var. *nudicaulis* are about the same length on average as the most proximal (on avg. 1.2 x longer) whereas the most distal rank of prickles are almost twice as long on average (1.7 x longer) in var. *crassicaulis*.

Paratypes. USA, CALIFORNIA, **San Luis Obispo Co.**: Estero Bluffs State Park on west side of Hwy 1 upcoast from the town of Cayucos, [35.45425°, -120.953117°]. Scattered under *Baccharis pilularis* DC. stand at base of hill 1.13 road mi W of San Geronimo Crk, 8 Mar 2010, *Michael Walgren 85* (OBI). Nipomo Dunes Complex, riparian corridor along northern base of Big Coreopsis Hill. Along SW fringes of riparian corridor. Plants 3–7 dm tall with yellow corollas. 23 May 1993, *Arthur W. Hazebrook 228* (OBI). North of San Simeon along ridge just S of Arroyo de la Cruz [35.71083°, -121.27889° +-10000m]. Windswept coastal hills with severely wind-pruned shrubby vegetation; occasional in sheltered area on wind-shallow slopes, fls yellow. 11 Apr 1981, *David J. Keil 14367* (OBI,UCR) with *Rhonda Riggins, Dirk Walters, Malcolm McLeod, Melissa Luckow, et al.* S of Oso Flaco in stabilized area of Nipomo Dunes [35.03°, -120.6175° +-2000m. NAD83]. Growing in protected place at inner border of the dunes, Herbaceous plant with taproot; fls yellow. 29 Mar 1972, *Robert J. Rodin 8621* (OBI). Nipomo Mesa. Big Coreopsis Hill S of Oso Flaco Lake, at end of Oso Flaco Lake Rd, 3.6 mi from jctn with State Hwy 1 [35.03°, -120.6175° +-2000m. NAD83] Locally common with *Populus trichocarpa* Hook. 10 Mar 1984, *David J. Keil 18088* (OBI) with *K. Jones, S. Cochrane, M.G. McLeod, Bernard and Lela Burdette, Austin Griffiths*. **Santa Cruz Co.**: approx. 17 mi NW of Santa Cruz (between California 1 (Cabrillo Hwy) and Swanton Rd, SE of Greyhound Rock, approx. 0.4 km E of old Seaside School. on SW side of Scott Creek); H-H Ranch, Scott Creek [37.083332°, -122.25°], in riparian habitat along horse trail; in decomposed Santa Cruz mudstone enriched with humus; along ...., lvs yellow-green, bristles whitish. Mar 6 1983, *Roy E. Buck, James A. West 144* (UCJEPS). Approx. 17 mi NW of Santa Cruz (between California 1 (Cabrillo Hwy) and Swanton Rd, SE of Greyhound Rock, W and NW of old Seaside School. in "Magic Triangle"); H-H Ranch, "Magic Triangle [37.083332°, -122.25°], at N end of divide ridge; on NNE-facing slope in N coastal scrub; in decomposed mudstone; among ..., leaves yellow-green; bristles whitish. 06 March 1983, *Roy E. Buck, James A. West 149* (UCJEPS); United States, California, Santa Cruz, ca 17 mi. NW Santa Cruz (between CA 1 (Cabrillo Hwy) & Swanton Rd, SE of Greyhound Rock & W & NW of old Seaside School, Swanton, bottom of "Lasher Marsh Gulch" near mouth); H-H Ranch [37.083332°, -122.25°], in tall *Rhamnus* scrub; shade; soil from Santa Cruz mudstone enriched w/ humus, marginal bristles ... whitish; lvs. ... yellow-green. 03 April 1983, *Roy E. Buck, James A. West 177* (UCJEPS); United States, California, Santa Cruz, approx. 17 mi nw of Santa Cruz (just N of Swanton Rd, 1.0 km NW of old Seaside School, Swanton, H-H Ranch); H-H Ranch [37.083332°, -122.25°], on upper se-facing slope of small gulch draining into main gulch; in shade just inside forest; lvs yellow-green; bristles whitish. 27 March 1983, *Roy E. Buck, James A. West 172* (UCJEPS). Approx. 17 mi NW of Santa Cruz (between California 1 (Cabrillo Hwy) and Swanton Rd, SE of Greyhound Rock, W and NW of old Seaside School. mouth of Big Willow Gulch); H-H Ranch, Big Willow Gulch [37.083332°, -122.25°], growing in *Aesculus californica* (Spach) Nutt. grove. 22 January 1985, *James A. West 298* (UCJEPS).

Common Names

We recommend Pacific Fringed-snakeroot as the common name for *Sanicula crassicaulis* var. *nudicaulis,* referring to its characteristic fringed leaf margins. The common names Pacific Sanicle, Gambleweed, and Pacific Blacksnakeroot are in use for *Sanicula*

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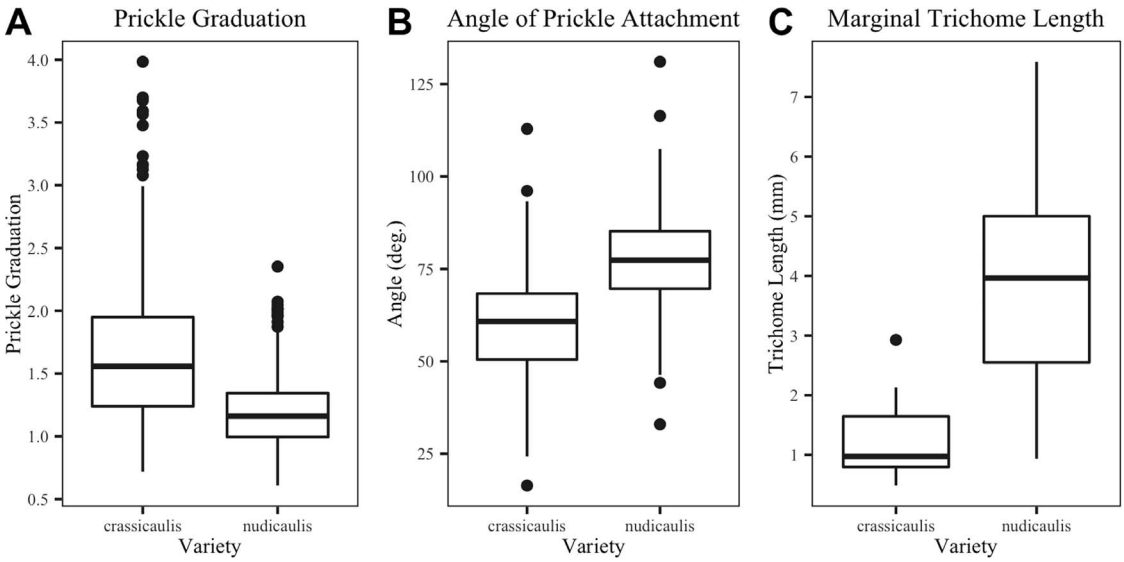

FIG. 4. Measured trait value distributions for var. *crassicaulis* and var. *nudicaulis*. (A) Prickle graduation. (B) Angle of prickle attachment to the schizocarp. (C) Leaf marginal trichome length. Angle of attachment of the prickle to the schizocarp is the average of three measured prickles per fruit (see Fig. 3 C and D). Prickle graduation is calculated as the length of the most distal prickle divided by the length of the most proximal.

crassicaulis var. *crassicaulis* and Pacific Blacksnakeroot for *Sanicula crassicaulis* var. *tripartita* (e.g., Calflora 2020; USDA Plants 2020), and here we recommend that they be restricted to vars. *crassicaulis* and *tripartita*.

RESULTS AND DISCUSSION

Trait Values

Our examination of the type specimens for *S. crassicaulis* var. *crassicaulis*, as well as other specimens from California and Chile, led us to conclude that *S. crassicaulis* var. *nudicaulis* can be distinguished from the type material by using the graduation of the prickles on the schizocarps, the angle of attachment of the prickles to the schizocarps and the length of the leaf marginal trichomes.

On average, the most proximal prickles on the schizocarps of var. *crassicaulis* are 1.7 times longer than the most distal (mean = 1.7, SD = 0.6, min = 0.7, max = 4.0) and the most proximal prickles on var. *nudicaulis* are 1.2 times longer than the most distal (mean = 1.2, SD = 0.3, min = 0.6, max = 2.4). Variety *nudicaulis* has on average a prickle attachment angle of 77° (mean = 77°, SD = 13°, min = 33°, max = 131°), 17° degrees greater than var. *crassicaulis* (mean = 60°, SD = 13°, min = 16°, max = 113°). On average, var. *crassicaulis* has leaf marginal trichomes that are 1.3 mm long (mean = 1.3, SD = 0.5, min = 0.5, max = 2.1) and var. *nudicaulis* has leaf marginal trichomes that are 4.0 mm long (mean = 4.0, SD = 1.5, min = 0.9, max = 7.6) (Fig. 4).

DISTRIBUTION AND ECOLOGY

The majority of collections made of *Sanicula crassicaulis* var. *nudicaulis* are from shaded areas, often appearing more mesic than the surrounding areas. It is unclear how strong this pattern may be. More collections and observations of this plant will allow for a more accurate understanding of its ecology. The authors have also observed *S. crassicaulis* var. *nudicaulis* to flower earlier than *S. crassicaulis* var. *crassicaulis*. Of interest is the fact that the authors have never observed var. *nudicaulis* growing more than about 100 meters away from var. *crassicaulis*. This supports the argument for designation of a unique variety in that we would expect to see intergradation where the two taxa are occurring so close together, but they remain morphologically distinct. Additionally, despite the fact that the two taxa are so commonly found close together, it is uncommon to find them actually growing side by side (but see Fig. 5). This suggests that perhaps there is some segregation between microclimates. This tendency to segregate along a moisture gradient as well as the potential for differential flowering times are two hypothesized barriers to reproduction. Nevertheless, this is extreme sympatry for sister taxa. Recent genomic work has revealed mechanisms whereby closely related species or ecotypes experience low levels of gene flow but maintain distinct phenotypes. Perhaps the most commonly observed mechanism at this time is the presence of large structural variants in the genome, such as inversions, that carry co-adapted gene complexes (Lamichhaney and Andersson 2019; Mérot 2020; Wellenreuther and Bernatchez, 2018). These prevent gene flow from

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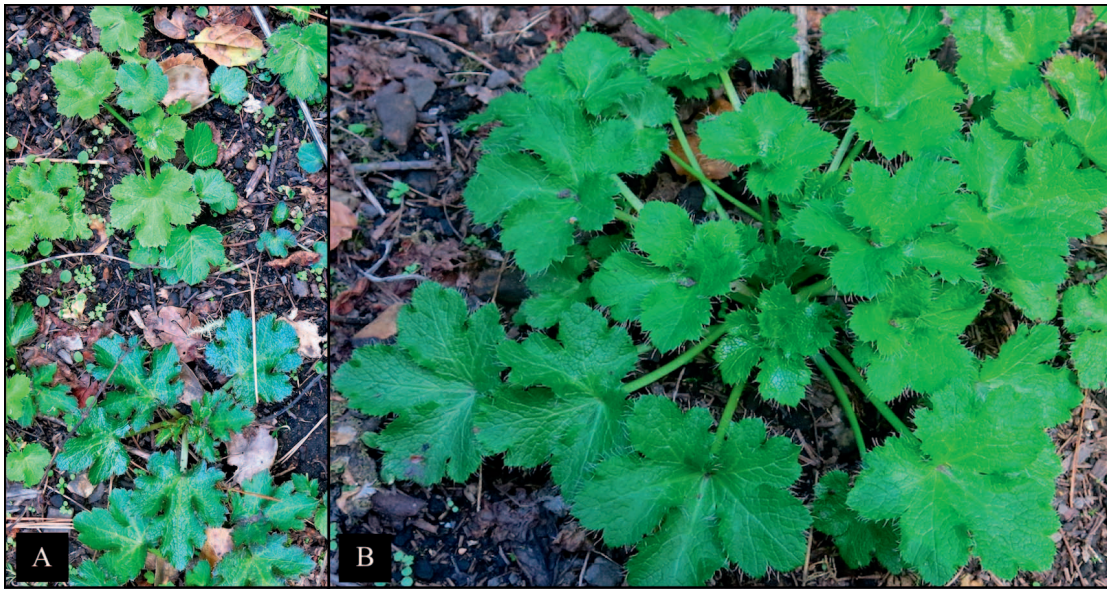

FIG. 5. Image (A) shows vars. *crassicaulis* (bottom) and *nudicaulis* (top) occurring together and Image (B) shows the typical stature and leaf morphology of var. *nudicaulis*. Both photos were taken at Jacks Peak, Monterey County, CA by Dylan Neubauer.

disrupting co-adapted gene complexes by limiting recombination in heterozygotes carrying different versions of the structural variant. Given the polyploid and polymorphic nature of the *S. crassicaulis*, structural variants within the genome are not inconceivable. This is supported by the fact that the only currently named variety in the species, *S. crassicaulis* var. *tripartita* is sympatric with var. *crassicaulis* and as far as we know, shows no strong habitat or flowering time divergence (Bell 1954). This suggests a high degree of post-mating reproductive isolation. Vars. *tripartita* and *nudicaulis* could represent stable ecotypes that experience low levels of gene flow but maintain divergent phenotypes. Crossing or genetic experiments would be valuable for determining the extent of reproductive isolation and the genetic basis for stable phenotypic divergence in the *Sanicula crassicaulis* complex.

POLYPLOIDY

Sanicula crassicaulis is known to have populations of tetraploid, hexaploid, and octoploid individuals. One potential explanation for the distinctness of *S. crassicaulis* var. *nudicaulis* is that it could represent population of previously unknown diploid individuals of *S. crassicaulis*. Ritchie Bell conducted an extensive project in 1954 to determine whether the ploidy levels in *S. crassicaulis* could be separated morphologically for the purpose of distinguishing separate taxa. He found no consistent morphological traits that could be used to determine the ploidy of an individual. In examining Bell's voucher specimens, we found one specimen (UC1714589) which appears to fall into var. *nudicaulis*. Bell marked on the label that it was tetraploid, leading us to conclude that var. *nudicaulis* is unlikely to represent any unique ploidy level in the species. However, this is not conclusive evidence and further analysis of the ploidy of var. *nudicaulis* could be enlightening.

Bell recognized one variety, *S. crassicaulis* var. *tripartita,* in his publication (Bell 1954). These plants are found primarily in Washington and are distinguished by their ternate leaves and purple-tinged flowers. All of these plants are octoploid and represent the only distinct polyploids that Bell found. However, there are octoploid individuals in California that do not fall into this taxon. This suggests that there is potential for differentiation within a ploidy level in the species. We believe this is one possibility to explain the distinct, putatively tetraploid, plants that we designate as var. *nudicaulis*.

Where vars. *crassicaulis* and *nudicaulis* might fall along the continuum of speciation is very difficult to say. We do not feel that we have evidence to designate these plants as separate species, but we feel that the rank of variety is useful and appropriate. We see two distinct and stable forms of *Sanicula crassicaulis* in California, but the strength of reproductive isolation is unclear. Naming these as distinct taxa allows for further study of what might be incipient speciation, the result of introgression, or complex interactions within a polyploid genome. Without a taxonomic rank, these forms are very unlikely to receive further investigation. By documenting this diversity, we increase the potential for gaining further knowledge.

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KEY TO THE VARIETIES OF *SANICULA CRASSICAULIS* IN CALIFORNIA

1. Leaf marginal trichomes 1-2(3) mm long and strait, angle of attachment of prickles on the schizocarps acute (~60°), most distal rank of prickles + twice as long as most proximal.
   2. Basal leaves + ternately lobed (the central lobe of the lower cauline and last formed basal leaves always elongated) with undulate margins, flowers showing varying amounts of purple pigmentation in the petals, plants of Oregon and Washington . . . . . *S. crassicaulis* var. *tripartita*
   2′ Basal leaves variously lobed in 3-5 parts, but not so deeply cut to be ternate, with the central lobe rarely much elongated. Flowers consistently yellow . . . . . *S. crassicaulis* var. *crassicaulis*

1′ Leaf marginal trichomes (2) 3–4 (7) mm long and sinuate, prickles on the schizocarps attached more or less perpendicularly (~80°), most distal rank of prickles + the same length on average as the most proximal . . . . . *S. crassicaulis* var. *nudicaulis*

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